



Project Title:

Logical Car Parking System

Submitted by:

Nisha Anwar 2022-CE-05

Saniha Zahid 2022-CE-08

Minahil Anjum 2022-CE-04

Arisha Kaleem 2022-CE-15

Submitted to:

SIR AFEEF OBAID

Course:

DIGITAL LOGIC DESIGN

Semester:

THIRD

Date of Submission:

18-12-2023

Department of Computer Engineering
University of Engineering and Technology, Lahore

Table of Contents

Abstract.....	3
1. Introduction.....	3
2. Problem statement.....	3
3. Explanation.....	3
3.1. Components to be used.....	3
3.2. Block Diagram.....	3
3.3. Circuit Diagram.....	3
4. Methodology.....	4
5. Cost Analysis.....	4
6. Expected Output and post analysis of the system.....	4
7. Applications and advantages of the proposed system.....	4
References.....	4



ABSTRACT

Due to the proliferation in the number of vehicles on the road, traffic problems are bound to exist. The Logical Car Parking System is a digital logic design project aimed at creating an efficient and user-friendly car parking system. The project involves the use of digital logic gates, sensors etc. to manage parking spaces, guide drivers to available spots, and ensure safe and organized parking within a parking facility. This system will contribute to reducing traffic congestion and enhancing the overall parking experience.

[1]



INTRODUCTION

The escalating urbanization and increasing vehicular density in cities have led to a critical need for efficient car parking solutions. Conventional parking systems often fall short in meeting the demands of modern urban living, resulting in congestion, frustration, and environmental concerns. This project introduces a logical car parking system employing digital logic components such as AND gates, XOR gates, NOT gates, LEDs, IR sensors, and a battery. The system aims to optimize parking space utilization, alleviate congestion, and enhance the overall parking experience.

SOLUTION TO PROBLEM

The proposed system integrates digital logic design principles to create an intelligent parking management system. Infrared (IR) sensors are strategically placed in parking spaces to detect vehicle presence, while AND gates, XOR gates, and NOT gates are employed for logical decision-making. LEDs provide visual feedback, and a battery ensures continuous functionality, even in the absence of external power.

EXPLANATION

A logical car parking system is used to efficiently manage parking spaces in various locations, including commercial lots, residential complexes, and urban areas.

7408 IC (AND Gates): These digital logic components will be used to combine input signals, such as sensor data and user inputs, to make determinations regarding parking space availability and control.

7486 IC (XOR Gate): The XOR gate will be employed for exclusive decision-making scenarios, aiding in parking space allocation decisions.

7404 IC (NOT Gate): NOT gates are utilized to invert signals, indicating the availability or occupancy status of a parking space.

Battery 5V: The 5V batteries will provide power to all system components,



ensuring consistent operation.

IR Sensors: These sensors, located at each parking space, will detect the presence of vehicles. When a vehicle is detected, the sensor will send a signal to the control logic.

LEDs: LEDs of different colors will be used to indicate the status of parking spaces. Red LEDs will signify occupied spaces, green LEDs will denote available spaces, and

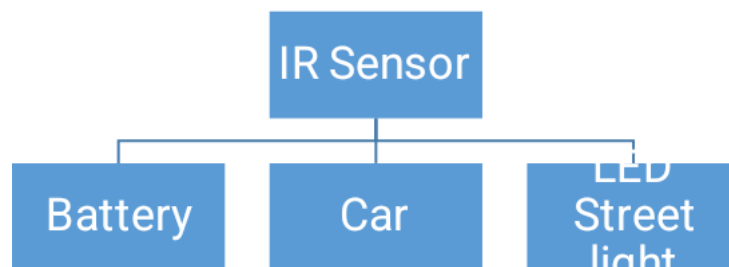
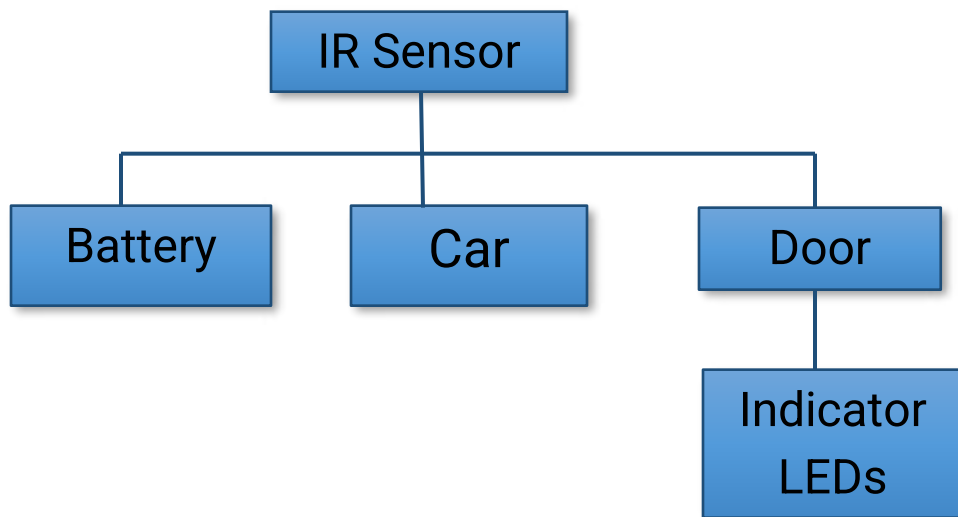
Yellow and green LEDs in larger circuit signify the number of cars entering the free space. White LEDs are being used in street light circuit and they will glow when a certain car will pass through them.

COMPONENTS USED

Sr. No	Component	Value	Quantity
1.	7408 IC (AND gate)	-	3
2.	7486 IC (XOR gate)	-	1
3.	7404 IC (Not gate)	-	2
4.	Battery	5V	3
5.	IR sensors	-	6
6.	LED Red	-	1
7.	LED Green	-	2
8.	LED yellow	-	3
9.	LED White	-	2
10.	Breadboard	-	2

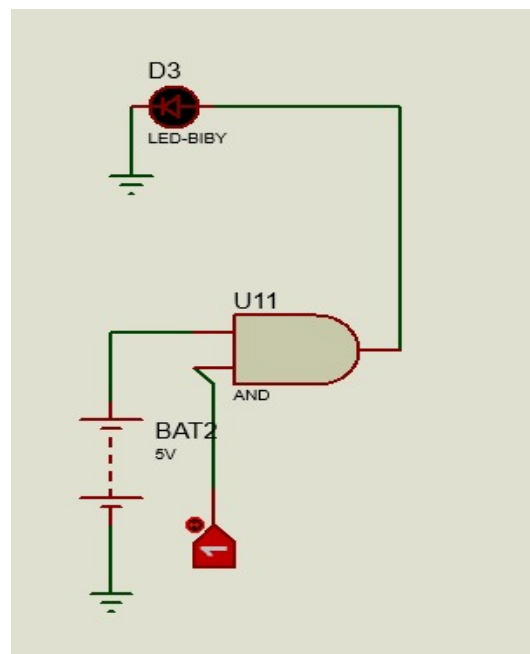
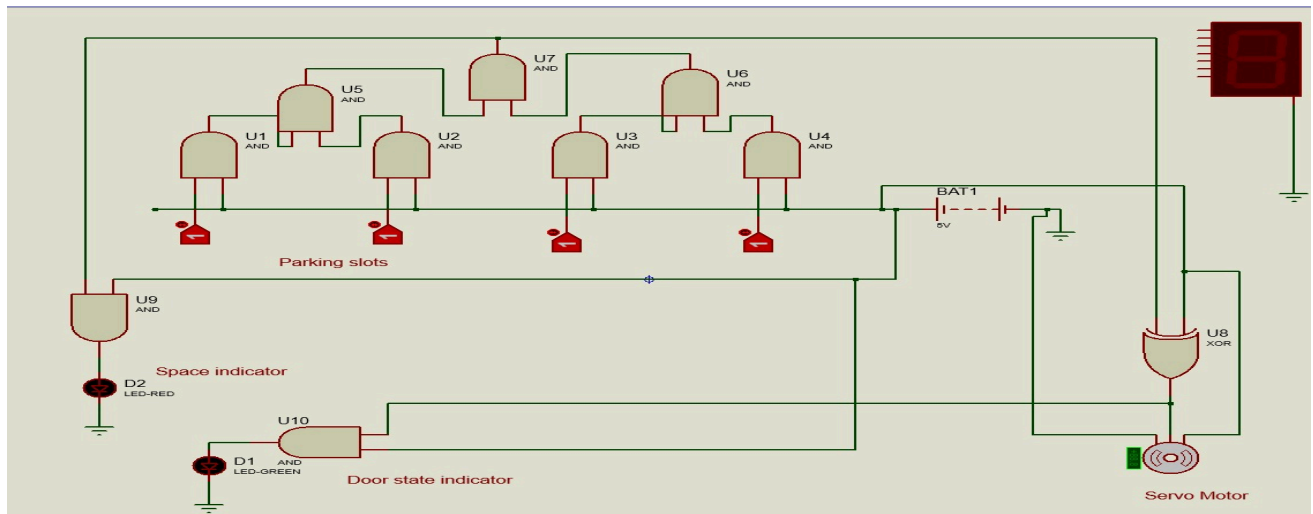


BLOCK DIAGRAM



CIRCUIT DIAGRAM





METHODOLOGY

1. Sensor Integration:

Install IR sensors at each parking space to detect the presence of vehicles. These sensors will be connected to the input of the digital logic system.

2. Logical Decision-Making:

AND gates process signals from multiple sensors to confirm vehicle presence.

XOR gates manage spot reservations, allowing only one valid reservation at a time.



NOT gates invert signals to indicate space availability or occupancy.

3. Control and Barrier System:

Implement the control logic using digital components. The system will make decisions based on the logic gates' outputs to manage barriers or gates for entry and exit points of the parking facility.

4. LED Displays:

Integrate LED displays (red, green, and yellow) to indicate parking space availability and provide directional guidance to drivers. Red LEDs indicate occupied spaces, green LEDs denote available spaces, and yellow LEDs will glow one by one with the passing of each vehicle.

5. Testing and Debugging:

Rigorously test the system to ensure that it accurately detects vehicle presence, makes real-time decisions, and operates reliably. This step is crucial to ensure that the system functions as intended.

6. User Interaction:

Users can interact with the system through a user-friendly interface to reserve parking spaces or check availability.

Conclusion:

This Digital Logic-Based Intelligent Car Parking System represents a significant step towards addressing the inefficiencies of traditional parking systems. By incorporating digital logic components, the system offers an intelligent, user-friendly, and sustainable solution for managing parking in urban environments.

TOTAL PROJECT COST

Sr.no	Components	Cost
1.	7408 IC (AND gate)	Rs.40 per IC
2.	7486 IC (XOR gate)	Rs.45
3.	7404 IC (NOT Gate)	Rs.40
4.	Battery	Rs.100
5.	IR sensors	Rs.150 per sensor



6.	LED Red	Rs.1
7.	LED Green	Rs.1
8.	LED Yellow	Rs.1
9.	LED White	Rs.1
9.	Breadboard	Rs.200

So the cost of "Logical car parking system is around 3000 including the expenses of other miscellaneous items used in the making of project.

Output of the system:

The output of the Digital Logic Car Parking System is to create an efficient parking management solution with the following key features:

Vehicle Presence: When a vehicle is detected by the IR sensors, the corresponding AND gates confirm its presence. The LED associated with the parking space turns ON, indicating occupancy.

Reservation Process: Users can reserve a parking space through the system's interface. The XOR gate ensures that only one reservation is valid at a time. The LED blinks or changes color to indicate a reserved space.

Availability Status: The NOT gates provide real-time information on the availability status of parking spaces. LEDs change color or blink to indicate whether a space is available or occupied.

Battery Backup: In the event of a power outage, the system seamlessly transitions to battery power. LEDs and reservation status remain functional, ensuring continuous service.

User Feedback: Users receive immediate visual feedback through the LEDs and the system interface. Clear indicators help users make informed decisions regarding parking space availability and reservations.

Applications and advantages of the system

APPLICATIONS

The Digital Logic Car Parking System has a broad range of applications, including:

Urban parking facilities

Commercial parking lots

Residential parking complexes

Institutional campuses

Any environment with parking management needs [1]

ADVANTAGES

Efficient Space Utilization:

Optimizes parking space usage, reducing the need for large parking areas.

Traffic Reduction:

Guides drivers to available spaces, minimizing congestion.

Enhanced Safety:

Reduces the risk of accidents and unauthorized access.

User-Friendly:

Provides real-time information to drivers, improving their parking experience.

Cost Savings:

Reduces the need for manual parking attendants and maintenance. [1]

References

1. <https://chat.openai.com/c/fb5d6851-28be-4566-beff-cbb61ffa4506>
2. <https://www.studocu.com/row/document/comsats-university-islamabad/digital-logic-design/dld-project-car-parking-system/25074052>
3. https://youtu.be/CqY_1PCJ4kU?si=VkiRQvmVR4Rx_WXH

